

3 A program has been written to check the value of a measurement. The measurement must be a positive number and given to three decimal places, for example, 3.982

- (a) (i) State suitable examples of normal and erroneous test data that could be used to test this program. For each example give the reason for your choice of test data.

Normal test data example

Reason

.....

Erroneous test data example

Reason

.....

[4]

- (ii) Explain why two pieces of boundary test data are required for this program. Give an example of each piece of boundary test data.

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[3]

- (b) Explain why verification is needed and how verification could be performed by this program.

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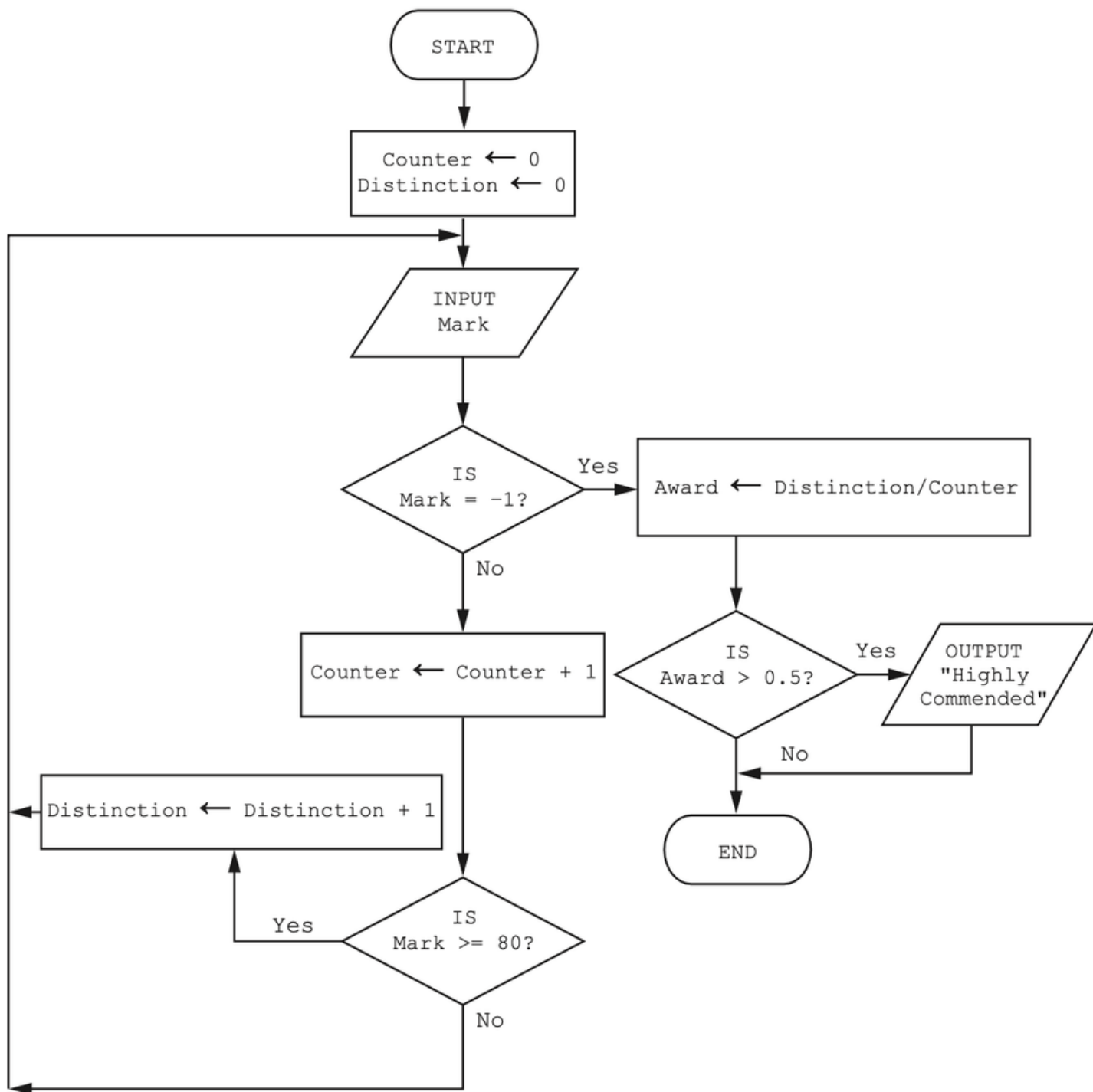
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[3]

- 4 The algorithm shown by this flowchart allows the input of examination marks for a class of students. A mark of -1 ends the process. If a mark is 80 or over then a distinction grade is awarded. The number of distinctions for the whole class is calculated. If this is over 50% of the class, the class is awarded a highly commended certificate.



Complete a trace table for the algorithm using this input data:
88, 74, 60, 90, 84, 87, 95, 72, 84, 66, −1

Counter	Distinction	Mark	Award	OUTPUT

NOVEMBER 2021 (2)

- 2 Tick (✓) one box in each row to identify if the statement is about validation, verification or neither.

Statement	Validation (✓)	Verification (✓)	Neither (✓)
a check where data is re-entered to make sure no errors have been introduced during data entry			
an automatic check to make sure the data entered has the correct number of characters			
a check to make sure the data entered is sensible			
a check to make sure the data entered is correct			

[3]

- 3 A program checks that the data entered is between 1 and 100 inclusive.

Identify **one** piece of normal, extreme and erroneous test data for this program, and give a reason for each.

Normal test data

Reason

.....

.....

Extreme test data

Reason

.....

.....

Erroneous test data

Reason

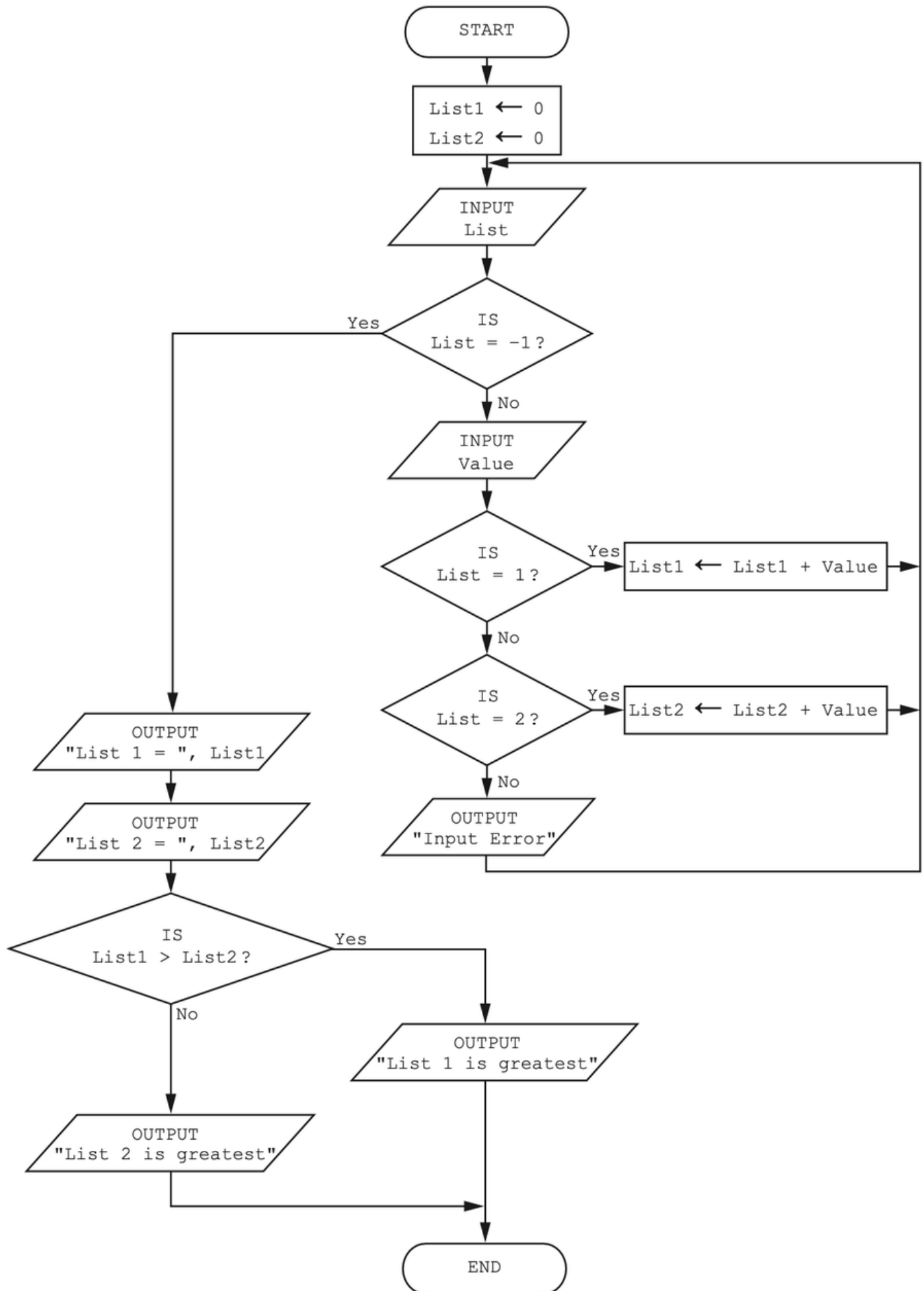
.....

.....

[6]

5 The flowchart represents an algorithm.

The algorithm will terminate if -1 is entered at the List input.



2, 77, 2, 16, 1, 35, 2, -7, 5, 18, 1, 11, 1, 12, 2, 20, -1, 18

[5]

3 A program has been written to check the length and content of a password. The password must be eight or more characters long and contain at least one special character, for example, `Secret!*!`

- (a) (i) State suitable examples of normal and erroneous test data that could be used to test this program. For each example give the reason for your choice of test data.

Normal test data example

Reason

.....

Erroneous test data example

Reason

.....

[4]

- (ii) Explain why two pieces of boundary test data are required for this program. Give an example of each piece of boundary test data.

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[3]

- (b) Describe **two** methods of verification that could be used to verify this data as it is input.

Method 1

.....

.....

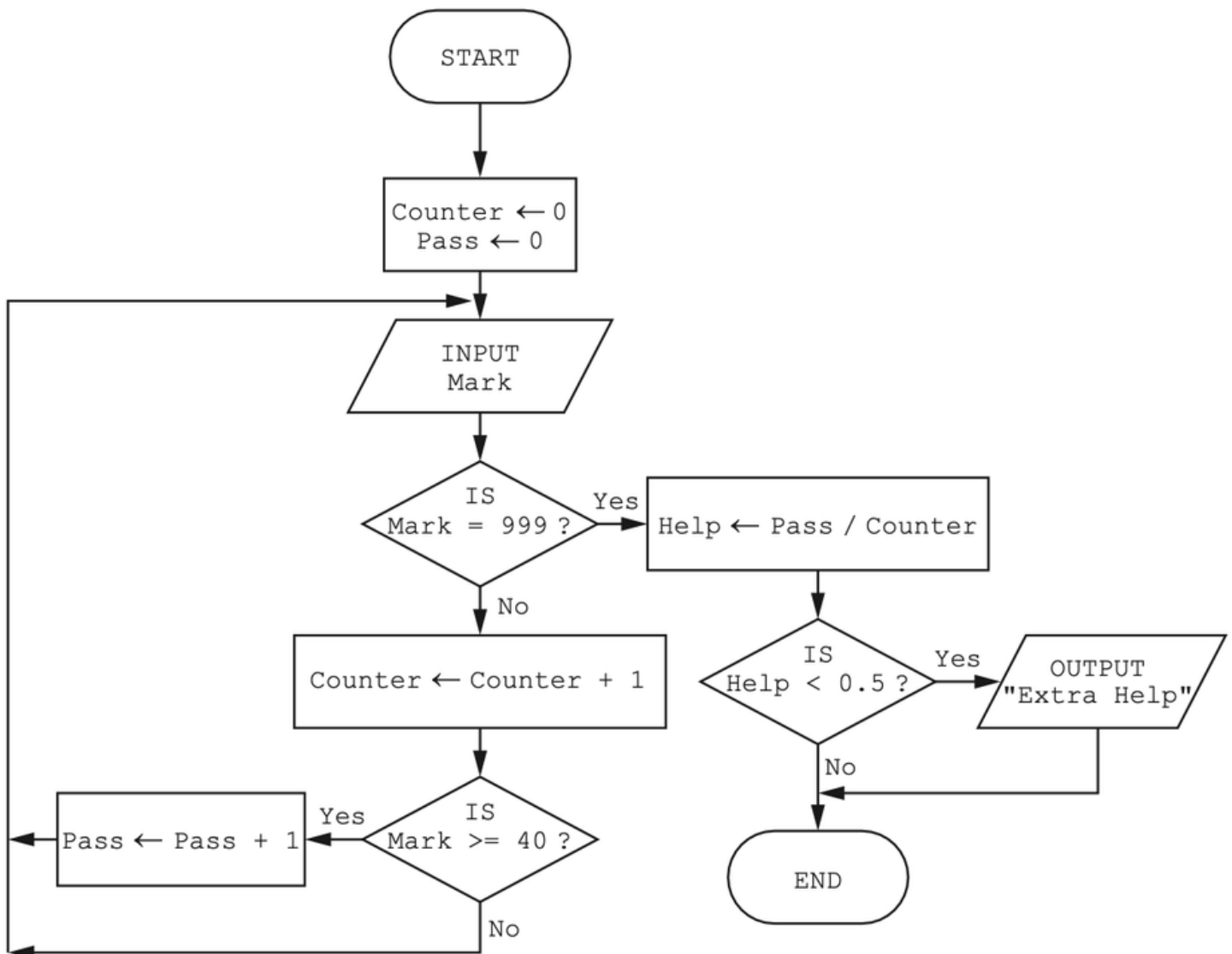
Method 2

.....

.....

[4]

- 4 The algorithm, shown by this flowchart, allows the input of examination marks for a class of students. A mark of 999 ends the process. If a mark is 40 or over then a pass grade is awarded. The number of pass grades is calculated for the whole class. If this is under 50% of the class, the class is offered extra help.



Complete a trace table for the algorithm using this input data:
88, 24, 60, 30, 44, 17, 25, 22, 54, 6, 999, -1

Counter	Pass	Mark	Help	OUTPUT

3 Four validation checks and five descriptions are shown.

Draw a line from each validation check to the most appropriate description.

Validation check

Description

length check

checks that the data input is
between two values

check digit

checks that the data input is
an integer

range check

checks that the data input has
three digits

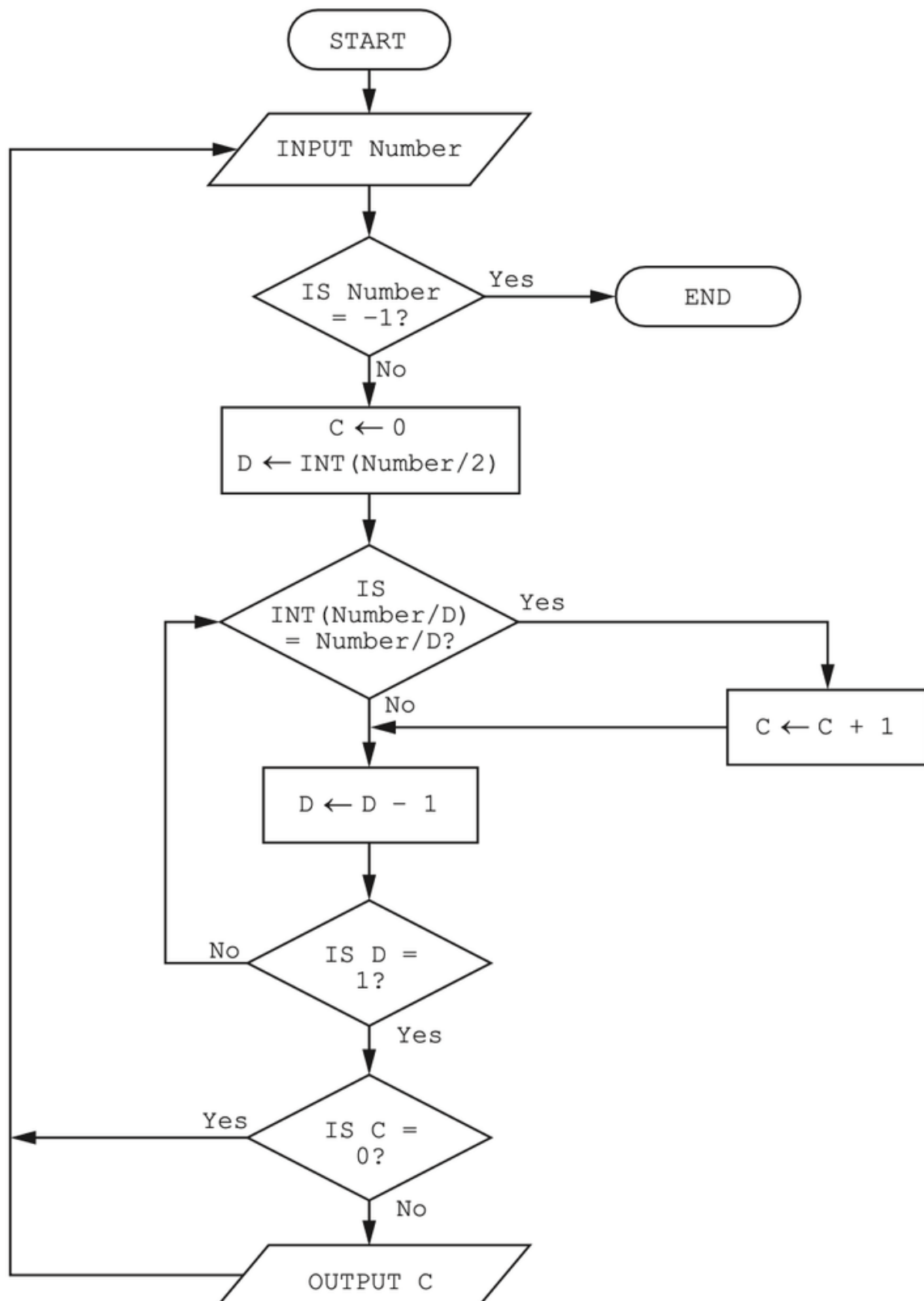
type check

checks that the data has
been input

checks that the data input has
the correct digits

4 This flowchart inputs a whole number. The function `INT` returns the integer value of a number. For example, `INT (7.5)` is 7

An input of `-1` ends the routine.



(a) Complete the trace table for the given algorithm using this input data:
 7, 6, 5, 4, -1, 12, 34

Number	C	D	OUTPUT

[6]

(b) Describe the purpose of this algorithm.

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..... [2]

(c) (i) Describe the problem that occurs if a whole number smaller than 4 and **not** equal to -1 is input.

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.....

.....

..... [2]

(ii) Explain how to change the flowchart to prevent this problem occurring.

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..... [2]

- 2 Tick (✓) **one** box in each row to identify the most appropriate data type for each description. Only **one** tick (✓) per column.

Description	Data type				
	Boolean	Char	Integer	Real	String
a single character from the keyboard					
multiple characters from the keyboard					
only one of two possible values					
only whole numbers					
any number					

[4]

- 3 Give **one** piece of normal test data and **one** piece of erroneous test data that could be used to validate the input of an email address.

State the reason for your choice in each case.

Normal test data

.....

Reason

.....

.....

.....

Erroneous test data

.....

Reason

.....

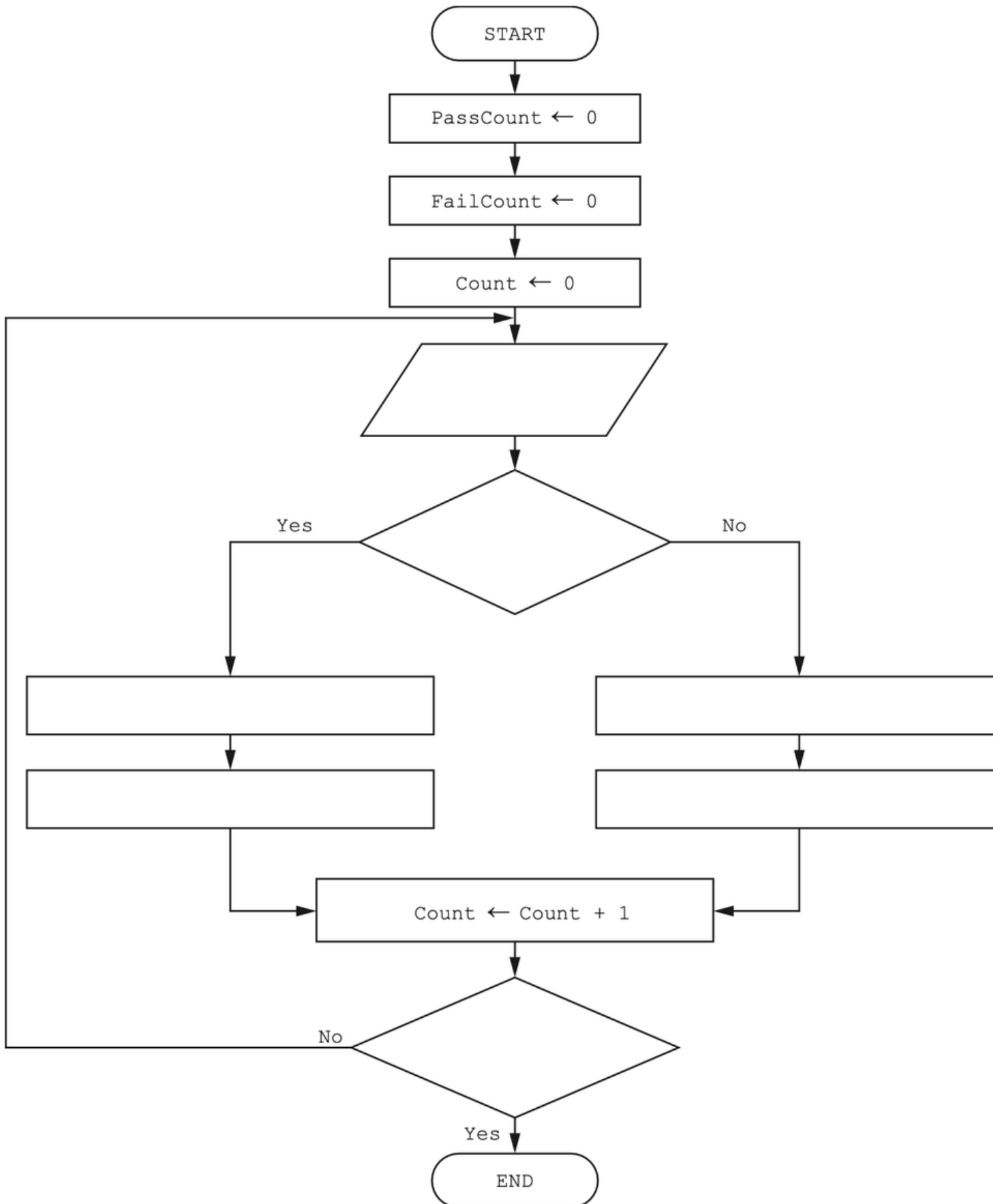
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[4]

- 4 The flowchart shows an algorithm that should allow 60 test results to be entered into the variable *Score*. Each test result is checked to see if it is 50 or more. If it is, the test result is assigned to the *Pass* array. Otherwise, it is assigned to the *Fail* array.

(a) Complete this flowchart:



5 The pseudocode represents an algorithm.

The pre-defined function **DIV** gives the value of the result of integer division.
For example, $Y = 9 \text{ DIV } 4$ gives the value $Y = 2$

The pre-defined function **MOD** gives the value of the remainder of integer division.
For example, $R = 9 \text{ MOD } 4$ gives the value $R = 1$

```
First ← 0
Last ← 0
INPUT Limit
FOR Counter ← 1 TO Limit
    INPUT Value
    IF Value >= 100
        THEN
            IF Value < 1000
                THEN
                    First ← Value DIV 100
                    Last ← Value MOD 10
                    IF First = Last
                        THEN
                            OUTPUT Value
                        ENDF
                    ENDF
                ENDF
            ENDF
        ENDF
    NEXT Counter
```

(a) Complete the trace table for the algorithm using this input data:

8, 66, 606, 6226, 8448, 642, 747, 77, 121

Counter	Value	First	Last	Limit	OUTPUT

[5]

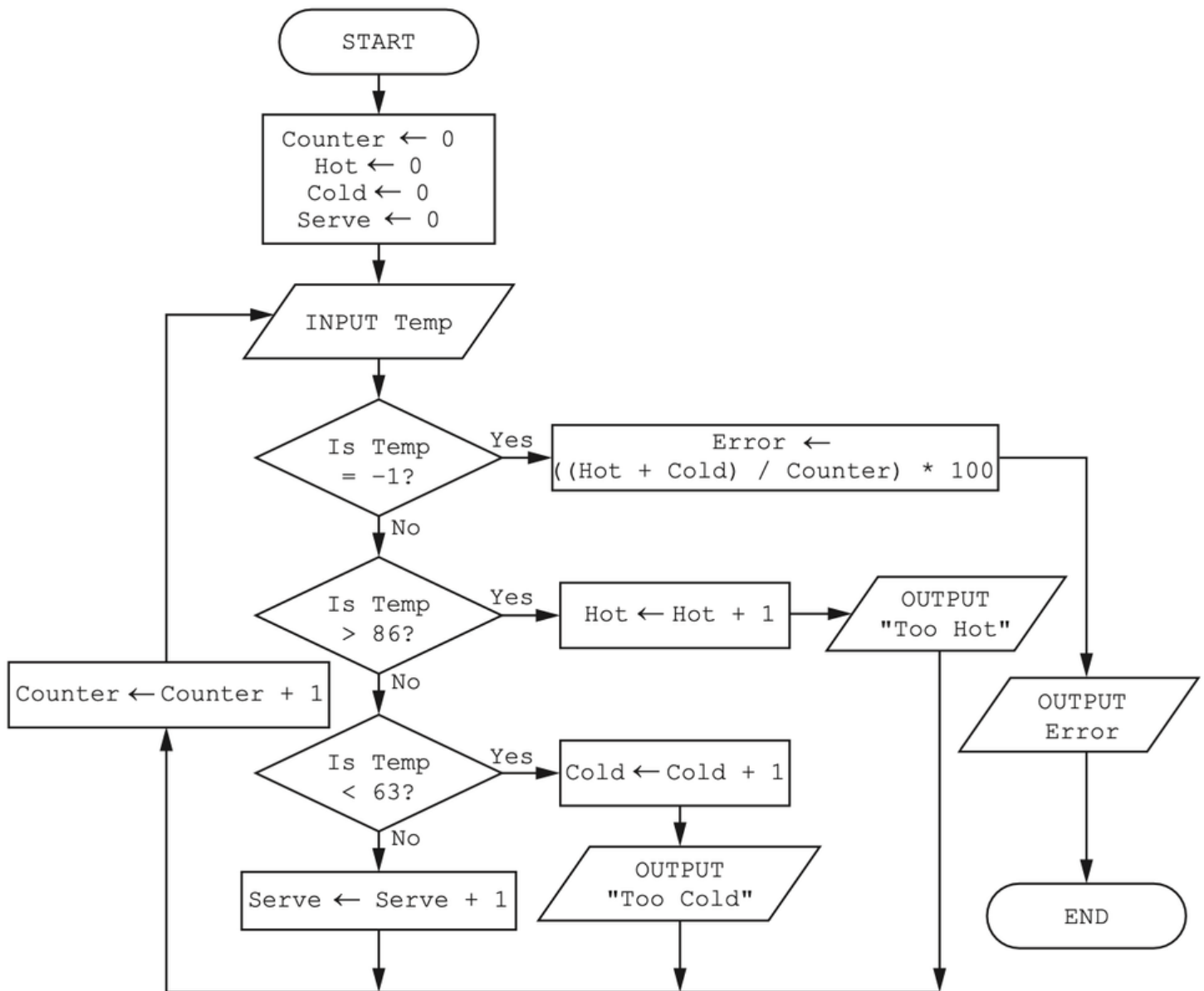
(b) Describe the purpose of the algorithm.

.....

.....

.....

4 This algorithm checks the temperature of hot food being served to customers.



(a) Complete the trace table for the algorithm using this input data:

75, 78, 84, 87, 91, 80, 75, 70, 65, 62, −1, 20

Counter	Hot	Cold	Serve	Temp	Error	OUTPUT

[7]

(b) State how the final output from the algorithm could be improved.

.....
..... [1]

(c) Identify the process in the algorithm that is **not** required.

.....
..... [1]

- 2 Tick (✓) **one** or more boxes in each row to match the type(s) of test data to each description.

Description	Types of test data			
	Boundary	Erroneous/ Abnormal	Extreme	Normal
test data that is always on the limit of acceptability				
test data that is either on the limit of acceptability or test data that is just outside the limit of acceptability				
test data that will always be rejected				
test data that is within the limits of acceptability				

[4]

- 3 Describe what is meant by the terms variable and constant and give an example of each in your answer.

Variable

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Constant

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[4]

4

The pre-defined function `DIV` gives the value of the result of integer division.

For example, $Y = 11 \text{ DIV } 4$ gives the value $Y = 2$

```

Count ← 0
INPUT Limit
FOR In ← 1 TO Limit
    Logic ← TRUE
    Test ← 2
    INPUT Number
    REPEAT
        IF Number / Test = Number DIV Test
            THEN
                Logic ← FALSE
            ELSE
                Test ← Test + 1
            ENDIF
    UNTIL NOT Logic OR Test >= Number DIV 2
    IF Logic
        THEN
            Store[Count] ← Number
            Count ← Count + 1
        ENDIF
NEXT In
FOR Out ← 0 TO Count - 1
    OUTPUT Store[Out]
NEXT Out

```

- (a) Complete the trace table for the algorithm using this input data:
5, 9, 5, 8, 10, 7

[illegible]

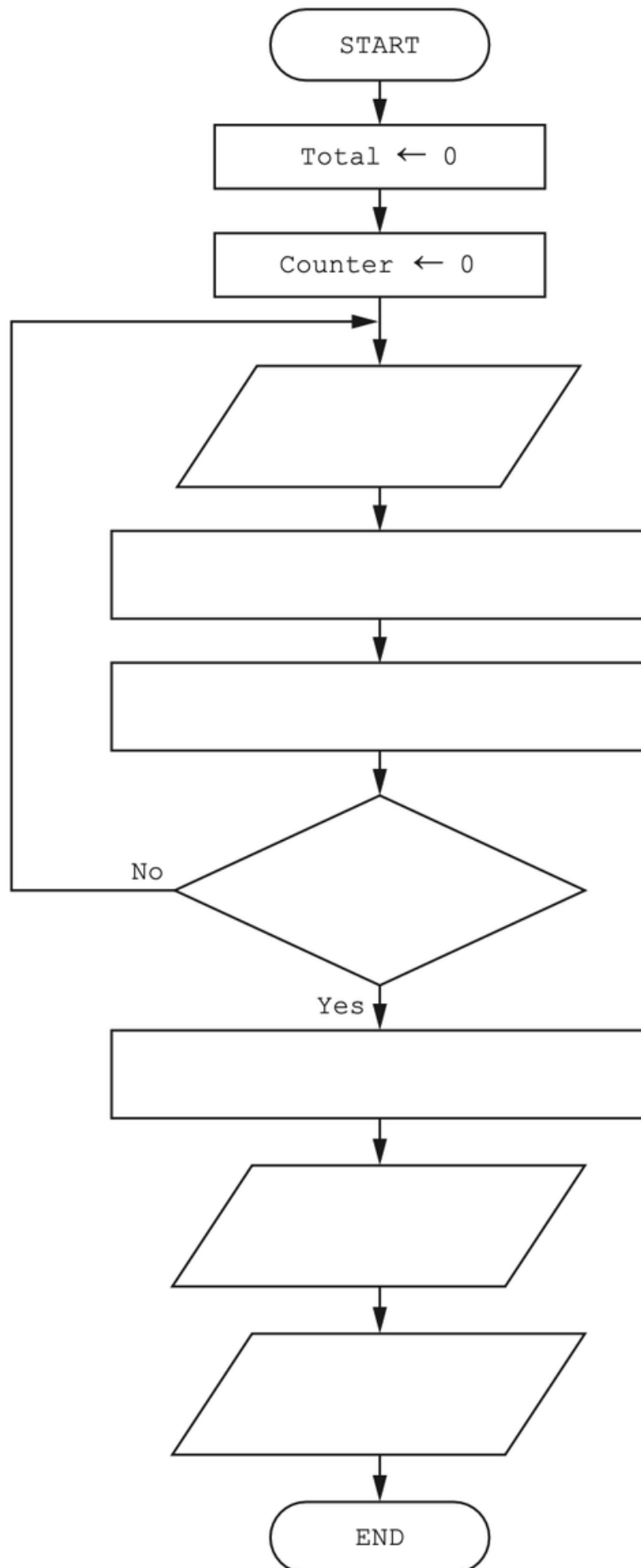
..... [2]

Describe how you could change this algorithm to make sure that only numbers that are 3 or greater are entered. Any pseudocode statements used in your answer must be fully described.

..... [3]

- 5 The flowchart shows an algorithm that should:
- allow 100 numbers to be entered into the variable `Number`
 - total the numbers as they are entered
 - output the total and average of the numbers after they have all been entered.

Complete this flowchart:



- 2 (a) Four descriptions and five pseudocode statements are shown.

Draw **one** line to link each description to its most appropriate pseudocode statement.
Not all pseudocode statements will be used.

Description	Pseudocode statement
a statement to count	FOR Count $\leftarrow$ 1 TO 10
a statement to total	Value $\leftarrow$ Value + NewValue
a statement to start a pre-condition loop	WHILE Value > 10 DO
a statement to start a post-condition loop	Value $\leftarrow$ Value + 1
	REPEAT

[4]

- (b) Write an algorithm in pseudocode, using a single loop, to output the average of 50 numbers that have been stored in the array `Number[]`

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..... [5]

3 Describe the purpose of test data. Include an example of a type of test data in your answer.

Description

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.....

.....

Example

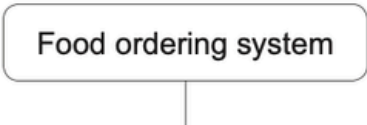
.....

[3]

5 A food ordering system is an example of a computer system that is made up of sub-systems.

- The food ordering system:
- allows the user to enter the details of the food they want to order and to pay for the order
 - displays food available as pictures or as a list.

Complete the structure diagram for the given parts of the food ordering system.



[4]

- 6 The energy efficiency of an electrical appliance is the percentage of useful energy out compared with the total energy in.

An algorithm has been written in pseudocode to calculate the energy efficiency of an appliance. Values for total energy in and useful energy out are input. The efficiency is calculated and output as a percentage.

The entry of the number -1 for either value stops the algorithm.

```
01 REPEAT
02     OUTPUT "Enter total energy in "
03     INPUT TotalEnergyIn
04     OUTPUT "Enter useful energy out "
05     OUTPUT UsefulEnergyOut
06     IF TotalEnergyIn <> -1 AND UsefulEnergy <> -1
07         THEN
08             Efficiency ← (UsefulEnergyOut / TotalEnergyIn) * 100
09             OUTPUT "Efficiency is ", Efficiency, "%"
10         ENDIF
11 UNTIL TotalEnergyIn <> -1 OR UsefulEnergyOut <> -1
```

- (a) Identify the **three** errors in the pseudocode and suggest corrections.

Error 1

Correction

.....

Error 2

Correction

.....

Error 3

Correction

.....

[3]

- (b) Write pseudocode to check for an efficiency of 92% or over for this appliance and to output "A-rated" if the efficiency is 92% or over.

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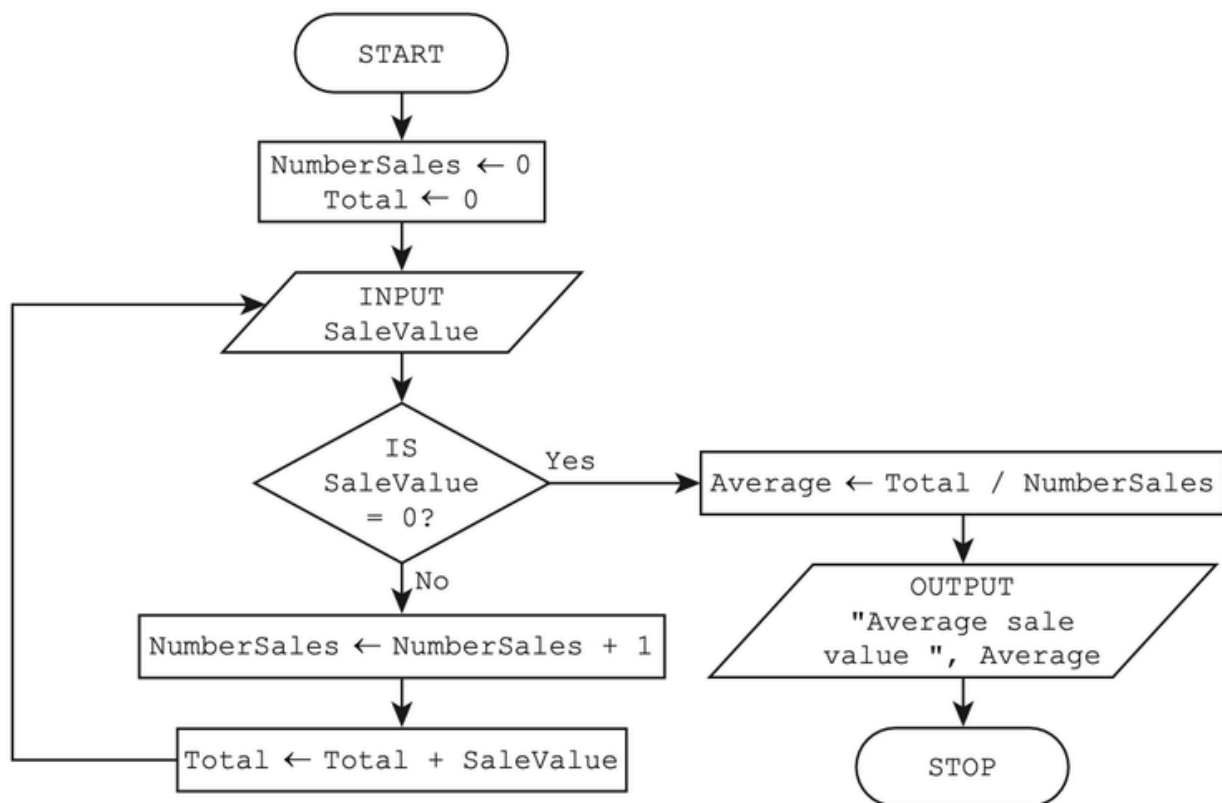
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..... [2]

8 This flowchart represents an algorithm to find the average value of a number of sales.



(a) Complete the trace table using this data:
5.50, 3.40, 6.25, 3.85, -11.00, 0

NumberSales	Total	SaleValue	Average	OUTPUT

[4]

(b) Identify the error in the algorithm and describe how to correct it.

Error

.....

Correction

.....

.....

- 1 (a) **Four** descriptions of stages in the program development life cycle are shown.

Draw **one** line to link each description to its most appropriate program development life cycle stage.

Not all program development life cycle stages will be used.

Program development life cycle description

Program development life cycle stage

develop an algorithm to solve the problem by using structure diagrams, flowcharts or pseudocode

analysis

detect and fix the errors in the program

coding

identify the problem and its requirements

design

write and implement the instructions to solve the problem

evaluation

testing

[4]

- (b) Identify **three** of the component parts after a problem has been decomposed.

- 1
-
- 2
-
- 3
-

[3]

3 (a) Describe what is meant by data validation.

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..... [2]

(b) A validation check is used to make sure that any value that is input is an integer between 30 and 200 inclusive.

Give **one** example of each type of test data to check that the validation check is working as intended. Each example of test data must be different.

Give a reason for each of your choices of test data.

Normal test data

Reason

.....

Abnormal test data

Reason

.....

Extreme test data

Reason

.....

[6]

- 5 An algorithm has been written in pseudocode to allow some numbers to be input. All the positive numbers that are input are totalled and this total is output at the end.
An input of 0 stops the algorithm.

```
01 Exit ← 1
02 WHILE Exit <> 0 DO
03     INPUT Number
04     IF Number < 0
05         THEN
06             Total ← Total + Number
07         ELSE
08             IF Number = 0
09                 THEN
10                     Exit ← 1
11             ENDIF
12         ENDIF
13 ENDIF
14 OUTPUT "The total value of your numbers is ", Number
```

- (a) Identify the **four** errors in the pseudocode and suggest a correction for each error.

Error 1

Correction

.....

Error 2

Correction

.....

Error 3

Correction

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Error 4

Correction

.....

[4]

- (b) Describe how you could change the corrected algorithm to record and output how many positive numbers have been included in the final total.

You do **not** need to rewrite the algorithm.

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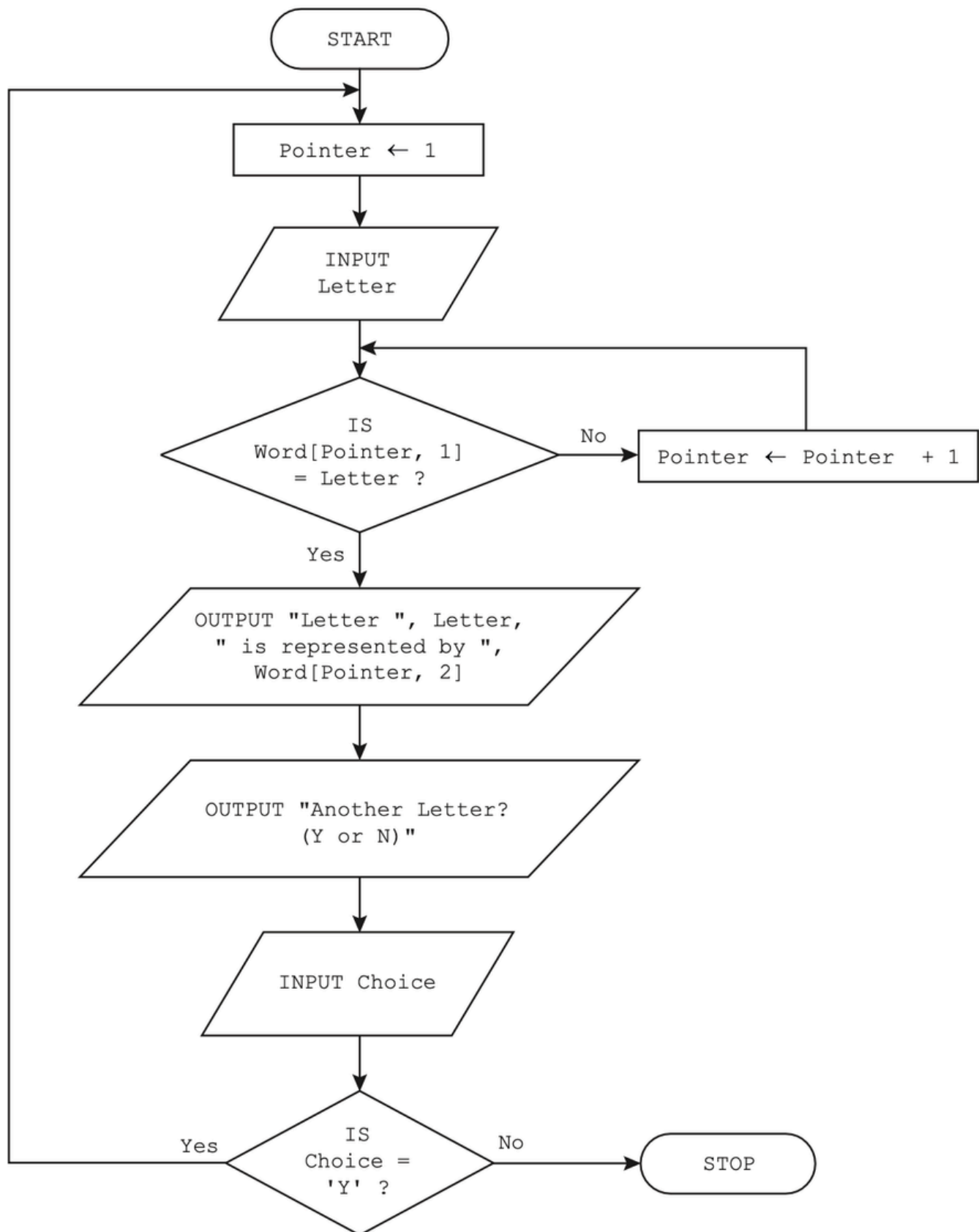
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7 The flowchart represents an algorithm.



The table represents the two-dimensional (2D) array `Word[]` which stores the first half of the phonetic alphabet used for radio transmission. For example, `Word[10,1]` is 'J'.

Index	1	2
1	A	Alpha
2	B	Bravo
3	C	Charlie
4	D	Delta
5	E	Echo
6	F	Foxtrot
7	G	Golf
8	H	Hotel
9	I	India
10	J	Juliet
11	K	Kilo
12	L	Lima
13	M	Mike

(a) Complete the trace table for the algorithm by using the input data: F, Y, D, N

Pointer	Letter	Choice	OUTPUT

(b) Identify the type of algorithm used.

.....
..... [1]

(c) Describe **one** problem that could occur with this algorithm if an invalid character was input.

.....
.....
.....
..... [2]

1 Tick (✓) **one** box to identify the first stage of the program development life cycle.

A Analysis

☐

B Coding

☐

C Design

☐

D Testing

☐

[1]

3 Identify **three** different ways that the design of a solution to a problem can be presented.

1

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2

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3

.....

[3]

- 4 A program needs to make sure the value input for a measurement meets the following rules:
- the value is a positive number
 - a value is always input
 - the value is less than 1000.

(a) Describe the validation checks that the programmer would need to use.

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..... [3]

(b) The program needs editing to include a double entry check for the value input.

(i) State why this check needs to be included.

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..... [1]

(ii) The input value needs to be stored in the variable `Measurement`
Write pseudocode to perform the double entry check until a successful input is made.

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..... [3]

- 7 An algorithm has been written in pseudocode to calculate a check digit for a four-digit number. The algorithm then outputs the five-digit number including the check digit. The algorithm stops when -1 is input as the fourth digit.

```
01 Flag ← FALSE
02 REPEAT
03     Total ← 0
04     FOR Counter ← 1 TO 4
05         OUTPUT "Enter a digit ", Counter
06         INPUT Number[Counter]
07         Total ← Total + Number * Counter
08         IF Number[Counter] = 0
09             THEN
10                 Flag ← TRUE
11         ENDIF
12     NEXT Counter
13     IF NOT Flag
14         THEN
15             Number[5] ← MOD(Total, 10)
16             FOR Counter ← 0 TO 5
17                 OUTPUT Number[Counter]
18             NEXT
19         ENDIF
20 UNTIL Flag
```

- (a) Give the line number(s) for the statements showing:

Totalling
Count-controlled loop
Post-condition loop

[3]

- (b) Identify the **three** errors in the pseudocode and suggest a correction for each error.

Error 1

Correction

.....

Error 2

Correction

.....

Error 3

Correction

.....

[3]

- (c) The algorithm does **not** check that each input is a single digit.
Identify the place in the algorithm where this check should occur.
Write pseudocode for this check.
Your pseudocode must make sure that the input is a single digit and checks for -1

Place in algorithm

Pseudocode

.....

.....

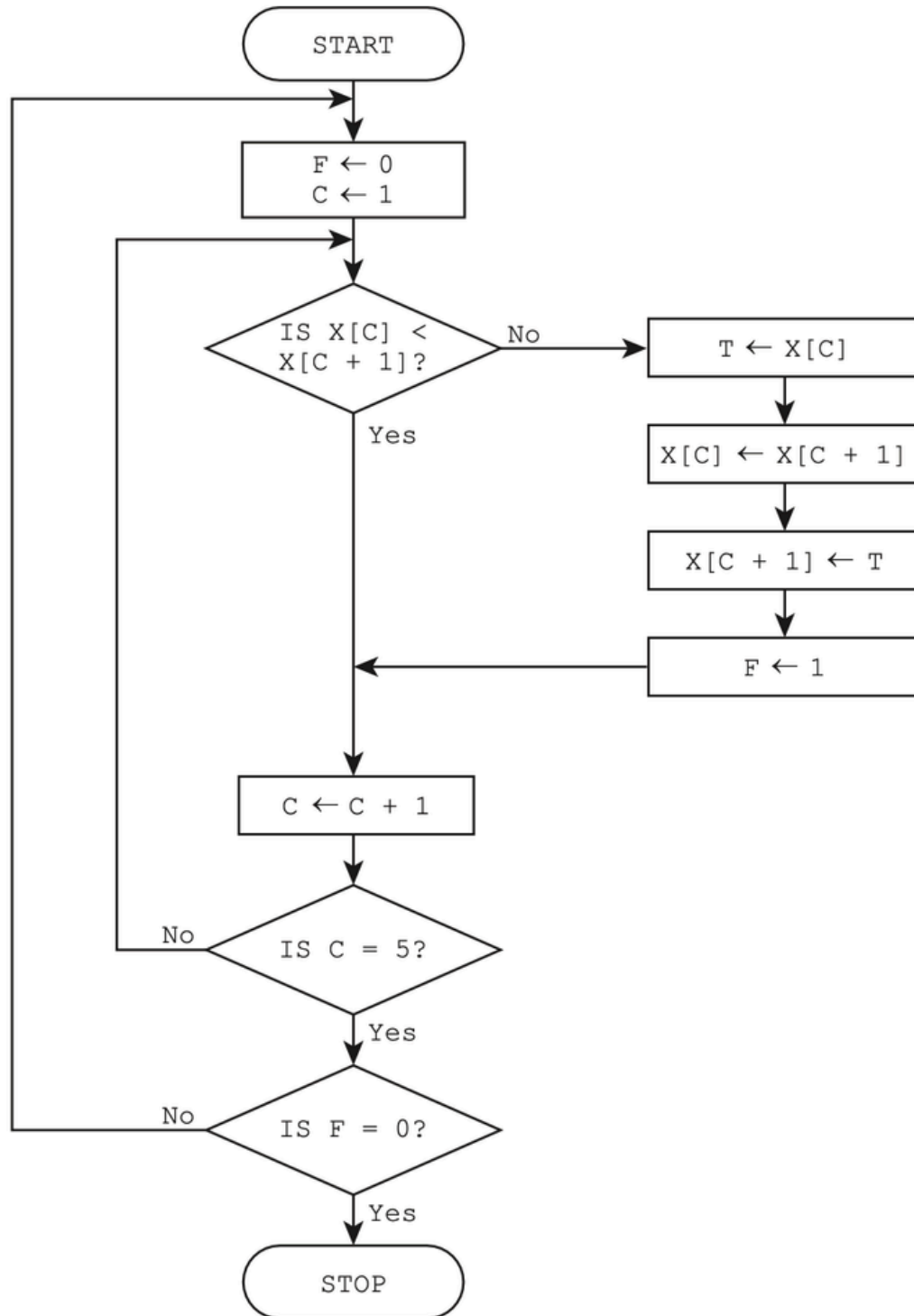
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[4]

9 This flowchart represents an algorithm.



(a) The array $X[1:5]$ used in the flowchart contains this data:

$X[1]$	$X[2]$	$X[3]$	$X[4]$	$X[5]$
10	1	5	7	11

Complete the trace table by using the data given in the array.

F	C	$X[1]$	$X[2]$	$X[3]$	$X[4]$	$X[5]$	T
		10	1	5	7	11	

[5]

(b) Describe what the algorithm represented by the flowchart is doing.

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..... [2]

- 4 (a) Explain why verification checks are used when data is input.

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..... [2]

- (b) Give **two** types of verification check and state how each one can be used.

Verification check 1

Use

.....

Verification check 2

Use

..... [4]

5 (a) Four descriptions of validation checks are shown.

Draw **one** line to link each description to the most appropriate check.

Not all checks will be used.

Description	Check
to check that the data entered is an integer	check digit
to check that some data has been entered	format check
to check that the data entered has an appropriate number of characters	length check
to check that an identification number contains no errors	presence check
	type check

[4]

(b) Write an algorithm in pseudocode to make sure that an input for the variable `Length` is between 15 and 35 inclusive. The code must iterate until a valid input has been made and the code must include appropriate messages.

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..... [3]

6 An algorithm has been written in pseudocode to allow 100 positive numbers to be input. The total and the average of the numbers are output.

```
01 Counter ← 100
02 Total ← 0
03 WHILE Counter > 100 DO
04     INPUT Number
05     IF Number > 0
06         THEN
07             Total ← Total + Counter
08             Counter ← Counter + 1
09     ENDCASE
10 ENDWHILE
11 OUTPUT "The total value of your numbers is ", Total
12 OUTPUT "The average value of your numbers is ", Total / 100
```

(a) Identify the **four** errors in the pseudocode and suggest corrections.

Error 1
Correction
.....

Error 2
Correction
.....

Error 3
Correction
.....

Error 4
Correction
.....

[4]

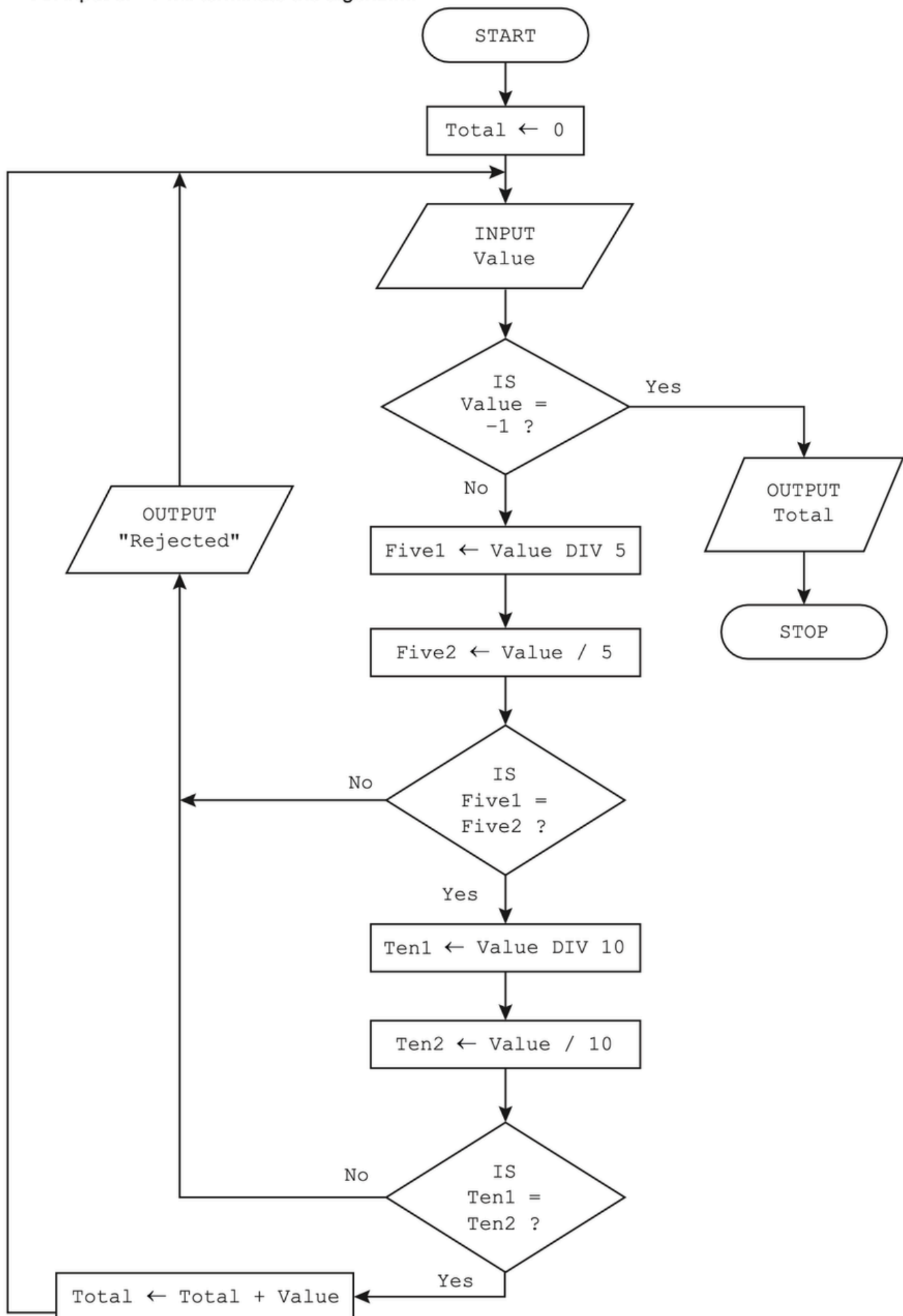
(b) Describe the changes you should make to the corrected algorithm so that a count-controlled loop is used to allow 100 positive numbers to be input.

You do **not** need to rewrite the algorithm.

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[5]

- 7 The flowchart represents an algorithm.
An input of -1 will terminate the algorithm.



(a) Complete the trace table for the input data:

5, 50, 52, 555, 57, 500, -1, 5500, 55

Total	Value	Five1	Five2	Ten1	Ten2	OUTPUT

[6]

(b) Describe the purpose of the algorithm.

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..... [2]

- 4 Circle the **three** words representing places where data may be stored.

array

constant

dimension

input

output

procedure

variable

[3]

- 5 The first stage of the program development life cycle is analysis. Two of the tasks in analysis are abstraction and decomposition.

- (a) Describe what is meant by abstraction.

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.....

.....

..... [2]

- (b) Identify **three** of the component parts when a problem has been decomposed at the analysis stage.

1

2

3 [3]

- (c) Identify and describe **one** other stage of the program development life cycle.

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..... [2]

6 An algorithm has been written in pseudocode.

```
01 DECLARE A[1:10] : STRING
02 DECLARE T : STRING
03 DECLARE C, L : INTEGER
04 L ← 10
05 FOR C ← 1 TO L
06     OUTPUT "Please enter name "
07     INPUT A[C]
08 NEXT C
09 FOR C ← 1 TO L
10     FOR L ← 1 TO 9
11         IF A[L] > A[L + 1]
12             THEN
13                 T ← A[L]
14                 A[L] ← A[L + 1]
15                 A[L + 1] ← T
16         ENDIF
17     NEXT L
18 NEXT C
19 FOR C ← 1 TO L
20     OUTPUT "Name ", C, " is ", A[C]
21 NEXT C
```

(a) State the purpose of this pseudocode algorithm.

.....
..... [1]

(b) State **four** processes in this algorithm.

1
.....
2
.....
3
.....
4
.....

[4]

(c) Meaningful identifiers have **not** been used in this algorithm.
Suggest suitable meaningful identifiers for:

The array:

A

The variables:

T

C

L

[3]

(d) State **two** other ways the algorithm can be made easier to understand and maintain.

1
.....
2
.....

[2]

- 8 A programmer is designing an algorithm to calculate the cost of a length of rope.
The program requirements are:
- input two values: the length of rope in metres `Length` and the cost of one metre `Cost`
 - perform a validation check on the length to ensure that the value is between 0.5 and 6.0 inclusive
 - calculate the price `Price`
 - output the price rounded to two decimal places.

Use the variable names given.

(a) State the name of the validation check.

..... [1]

(b) Complete the flowchart for this algorithm.



[6]

(c) Give **two** different sets of test data for this algorithm and state the purpose of each set.

Set 1

Purpose

.....

.....

Set 2

Purpose

.....

.....

[4]

(d) Complete the headings for the trace table to show a dry-run for this algorithm.
 You do **not** need to trace the algorithm.

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[3]

(e) Describe an improvement that should be made to the requirements for this algorithm.

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..... [2]

- 1 Tick (✓) **one** box to complete the sentence.

Verification is used to make sure that a value entered

A has **not** changed during input.

☐

B is an integer.

☐

C is correct.

☐

D is **not** a string.

☐

[1]

- 2 A type of validation check is a length check. Another type of validation check is used to make sure that any date entered is in the dd/mm/yyyy style:
dd means day, mm means month and yyyy means year.

(a) State the type of validation check used.

..... [1]

- (b) Give **one** example of normal test data and **one** example of abnormal test data you should use to make sure the check in part (a) is working properly.

State a reason for each of your choices of test data.

Normal

Reason

.....

Abnormal

Reason

.....

[4]

- (c) Describe how a length check could be used with the date entered.

.....

.....

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..... [2]

- 4 An algorithm has been written in pseudocode to allow the names of 50 cities and their countries to be entered and stored in a two-dimensional (2D) array. The contents of the array are then output.

```
01 DECLARE City ARRAY[1:50, 1:2] OF BOOLEAN
02 DECLARE Count : INTEGER
03 DECLARE Out : INTEGER
04 Count ← 1
05 IF
06     OUTPUT "Enter the name of the city"
07     INPUT City[Count, 2]
08     OUTPUT "Enter the name of the country"
09     INPUT City[Count, 2]
10     Count ← Count + 1
11 UNTIL Count = 50
12 FOR Out ← 1 TO 1
13     OUTPUT "The city ", City[Out, 1], " is in ", City[Out, 2]
14 NEXT Out
```

- (a) Identify the **four** errors in the pseudocode and suggest corrections.

Error 1

Correction

.....

Error 2

Correction

.....

Error 3

Correction

.....

Error 4

Correction

.....

[4]

- (b) Describe the changes you should make to the corrected algorithm to allow the name of a country to be input and to display only the stored cities from that country.

You do **not** need to rewrite the algorithm.

.....

.....

.....

.....

.....

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.....

.....

.....

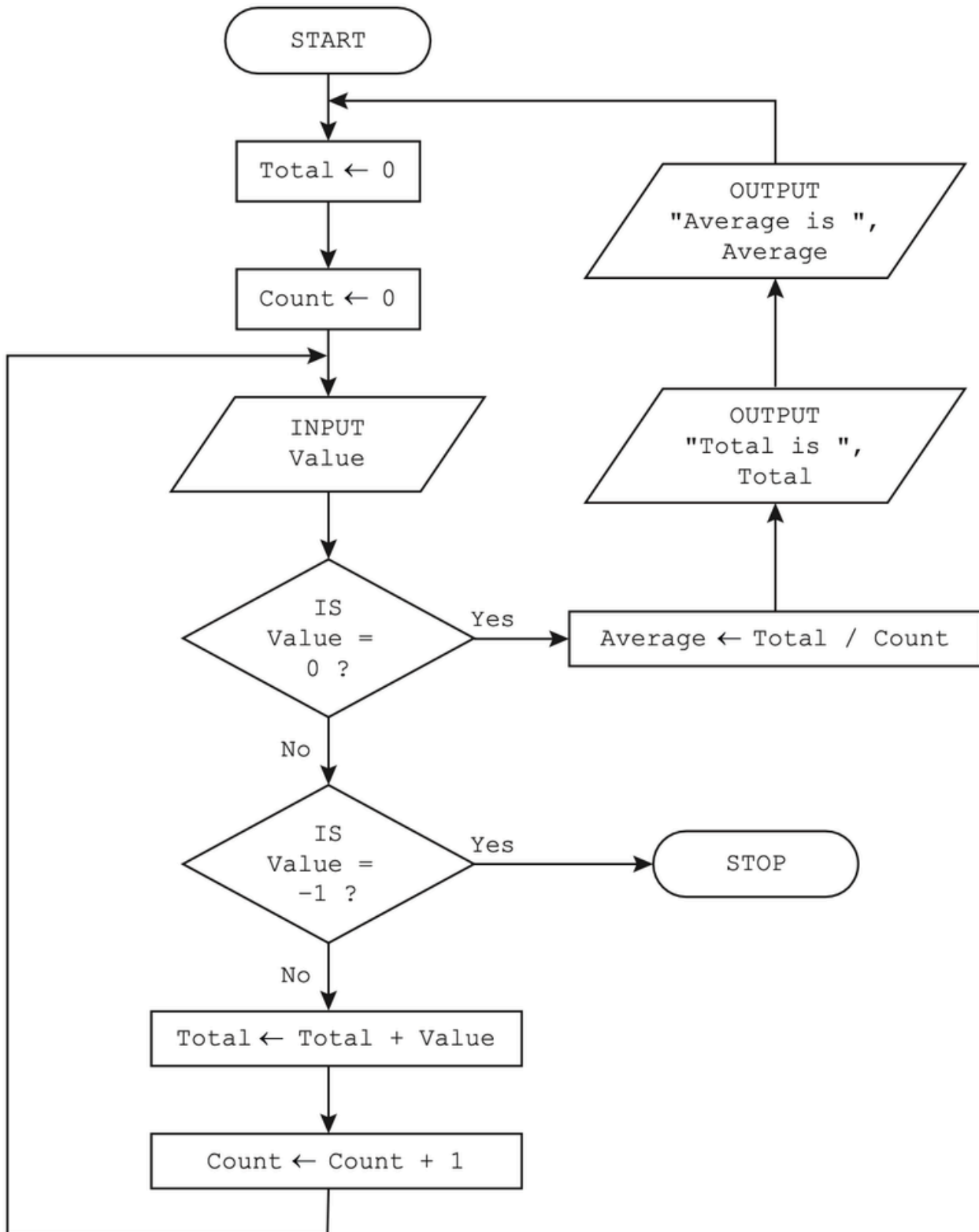
.....

.....

..... [5]

- 6 The flowchart represents an algorithm that performs a process on groups of values that are input. The algorithm will fail if the first value of any group is 0.

An input of -1 will terminate the algorithm.



(a) Complete the trace table for the input data:

25, 35, 3, 0, 57, 20, 25, 18, 0, -1, 307, 40, 0

Value	Average	Total	Count	OUTPUT

[5]

(b) Describe the purpose of the algorithm.

.....

.....

.....

..... [2]

1 Tick (✓) **one** box to show which term is an example of a verification check.

- | | | |
|----------|--------------------|--------------------------|
| A | Double entry check | ☐ |
| B | Format check | ☐ |
| C | Length check | ☐ |
| D | Presence check | ☐ |

[1]

3 (a) **Four** pseudocode descriptions and **five** pseudocode keywords are shown.

Draw **one** line to link each pseudocode description to the most appropriate pseudocode keyword. **Not** all pseudocode keywords will be used.

Pseudocode description	Pseudocode keyword
stores data in a file	OUTPUT
retrieves data from a file	WRITE
displays data on a screen	READ
enters data from a keyboard	OPEN
	INPUT

[4]

(b) Give **two** reasons for storing data in a file.

- 1
-
- 2
-

[2]

4 A programmer is writing a data entry program for booking theatre seats.
The programmer needs the program to accept only whole numbers that are greater than or equal to one and less than or equal to six.

(a) Give the names of **two** validation checks that are required for this program.

1
2 [2]

(b) Complete this pseudocode to perform your **two** validation checks, using your answers given in (a):

```
OUTPUT "Please enter the number of seats you want to book "
```



```
INPUT Seats
```

.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [5]

(c) Give **one** item of test data to use when testing this program.
State the reason for your choice of test data.

Test data
Reason for choice
..... [2]

- 5 An algorithm has been written in pseudocode to check if a new password is in a list of previously used passwords `OldList[]`
If the password is **not** found, the new password will be stored at the end of the list to replace "XXXX" already stored there.

```
01 OUTPUT "Enter your new password "  
02 INPUT NewPassword  
03 Posn ← 1  
04 Found ← FALSE  
05 REPEAT  
06     IF Password = OldList[Posn]  
07     THEN  
08         Found ← TRUE  
09     ELSE Posn ← Posn + 1  
10     ENDIF  
11 UNTIL Found AND OldList[Posn] = "XXXX"  
12 IF Found  
13 THEN  
14     OUTPUT "Password has been used before"  
15 ELSE  
16     INPUT "New password accepted"  
17     OldList[Posn] ← NewPassword  
18 ENDIF
```

(a) Identify the **three** errors in the pseudocode and suggest corrections.

Error 1

Correction

.....

Error 2

Correction

.....

Error 3

Correction

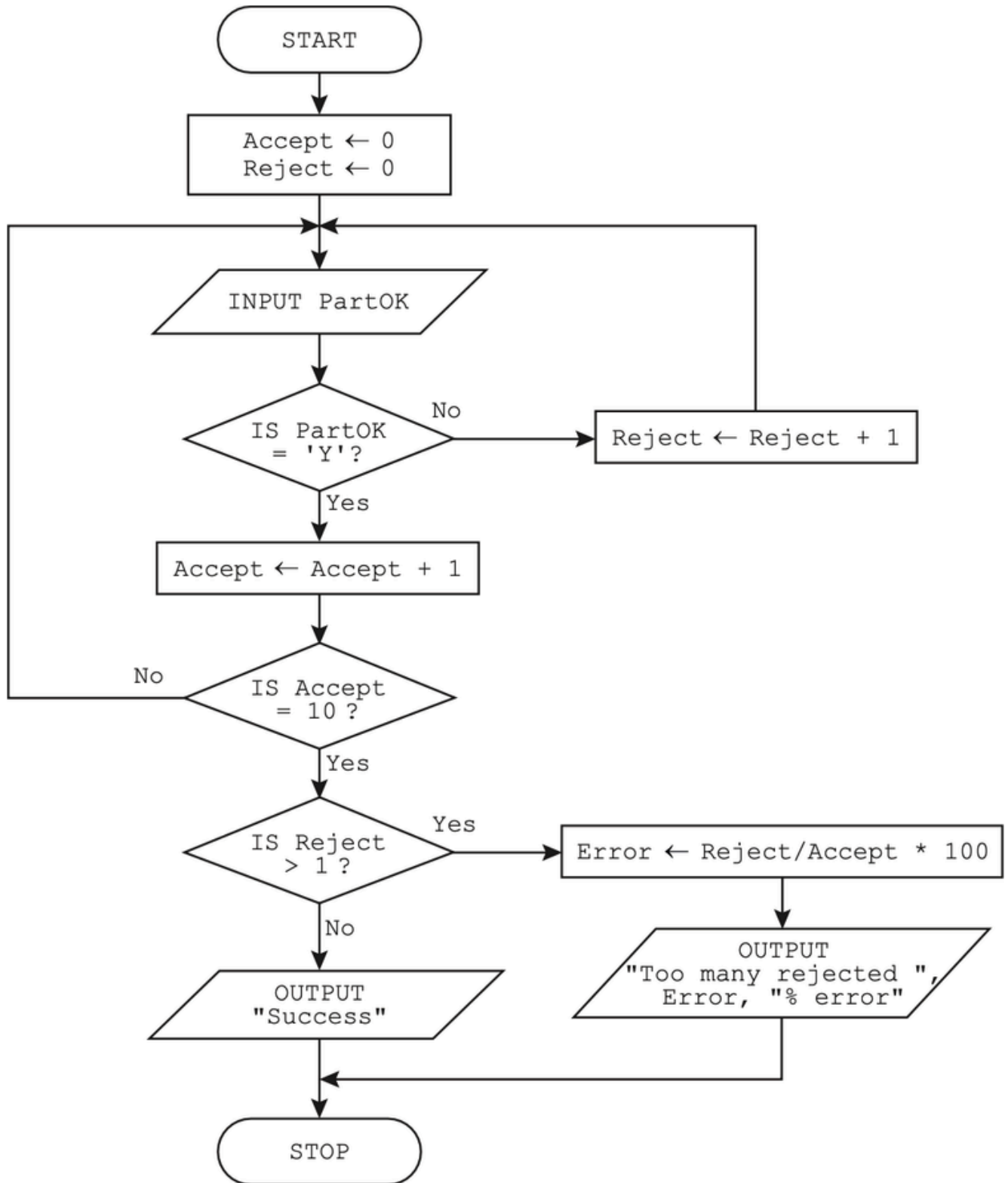
.....

[3]

(b) Complete this flowchart for the corrected algorithm:



8 This is an algorithm to find if a batch of parts has been manufactured successfully.



- Y, Y, Y, N, Y, Y, Y, Y, N, Y, Y, Y, Y

[illegible]

[5]

- (b)** Describe how the algorithm should be changed to accept 'Y' or 'y' for a successfully manufactured part.

[3]

[3]